

GabayPoz Recommender System — Version 2

Demonstration of Latent-Variable Program-First Recommendations

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Abstract

This document is an executable demonstration of the GabayPoz recommender, version 2. It is generated by running `demo.py` against the bundled sample fixtures under `data/sample/` and emitting a LaTeX fragment (`results.tex`) that this wrapper includes verbatim. Each subsection describes one synthetic student profile, including a v2 field-interest profile, Q7 strand, Q10 affordability, Q11 mobility, Q12 duration tolerance, Q13 professional-track aspiration, the top-three program recommendations, the primary school suggestion, and the rank-1 explanation and trace summary.

1. About this demonstration

GabayPoz is a constraint-aware, explainable college program recommender for Grade 11/12 students in Pozorrubio, Pangasinan. EDA established that tertiary access is shaped strongly by geography and household economics, so the recommender keeps Q11 commute and Q10 affordability as hard school filters. Version 2 separates interest and self-efficacy into 24 Likert items, uses Q7 as a small bounded context signal, scores against `program_profile_v2`, and emits recommendation trace rows for audit.

The current demo pipeline is:

1. **Student profile.** V2Q01–V2Q24 build interest and self-efficacy vectors; Q7 contributes only bounded context.
2. **Program ranking.** Programs are ranked first using student fit against `program_profile_v2`, municipality saturation as small context, Q12 duration penalty, and optional Q13 track boost.
3. **School selection.** For each selected program, feasible schools are filtered by Q11 commute and Q10 affordability.
4. **Persistence shape.** The model emits three ranked `model_recommendation` rows plus three trace rows using the v2 model ID; explanations and alternate schools remain in the API response.

The four profiles below exercise STEM, HUMSS, TVL, and GAS strand inputs.

2. Demonstrative profiles

D1: Health-oriented STEM strand, medicine aspiration, ready for board-exam programs

Model: `tds_recommender_v2`; Barangay id: 1; High field: `health`; Q7: STEM; Q10: C; Q11: C; Q12: A; Q13: A.

	Rank	Program	Primary school	Score
Top recommendations.	1	BS Nursing	PSU Lingayen	0.869
	2	BS Medical Technology	PSU Lingayen	0.815
	3	BS Computer Science	PSU Lingayen	0.352

Rank-1 explanation. BS Nursing matches your strongest Health signals. PSU Lingayen is the primary school suggestion because it passed your travel and affordability limits. Local Health field presence is included only as small context, not a job guarantee.

Rank-1 trace. Dominant field: **health**; track boost factor: 1.12; low-confidence reason: .

D2: Arts-oriented HUMSS strand, nearby-only travel

Model: tds_recommender_v2; Barangay id: 1; High field: **arts**; Q7: HUMSS; Q10: B; Q11: A; Q12: B; Q13: D.

	Rank	Program	Primary school	Score
Top recommendations.	1	AB Communication	PSU Lingayen	0.770
	2	BS Business Administration	PSU Lingayen	0.417
	3	Bachelor of Elementary Education	PSU Lingayen	0.315

Rank-1 explanation. AB Communication matches your strongest Arts signals. PSU Lingayen is the primary school suggestion because it passed your travel and affordability limits. Local Arts field presence is included only as small context, not a job guarantee.

Rank-1 trace. Dominant field: **arts**; track boost factor: 1.00; low-confidence reason: .

D3: Agriculture-oriented TVL strand, moderate budget

Model: tds_recommender_v2; Barangay id: 3; High field: **agriculture**; Q7: TVL; Q10: B; Q11: B; Q12: B; Q13: D.

	Rank	Program	Primary school	Score
Top recommendations.	1	BS Agriculture	PSU Lingayen	0.800
	2	BS Civil Engineering	PSU Lingayen	0.339
	3	BS Medical Technology	PSU Lingayen	0.301

Rank-1 explanation. BS Agriculture matches your strongest Agriculture signals. PSU Lingayen is the primary school suggestion because it passed your travel and affordability limits. Local Agriculture field presence is included only as small context, not a job guarantee.

Rank-1 trace. Dominant field: **agriculture**; track boost factor: 1.00; low-confidence reason: .

D4: STEM-oriented GAS strand, moderate budget

Model: tds_recommender_v2; Barangay id: 2; High field: **stem**; Q7: GAS; Q10: B; Q11: B; Q12: B; Q13: D.

	Rank	Program	Primary school	Score
Top recommendations.	1	BS Civil Engineering	PSU Lingayen	0.790
	2	BS Computer Science	PSU Lingayen	0.788
	3	BS Medical Technology	PSU Lingayen	0.533

Rank-1 explanation. BS Civil Engineering matches your strongest STEM signals. PSU Lingayen is the primary school suggestion because it passed your travel and affordability limits. Local STEM field presence is included only as small context, not a job guarantee.

Rank-1 trace. Dominant field: **stem**; track boost factor: 1.00; low-confidence reason: .

3. How to reproduce this PDF

```
cd demo
make                # runs demo.py then pdflatex twice
```

`make clean` removes the LaTeX intermediates and the generated `results.tex`; `make distclean` additionally removes `demo.pdf`. Re-running from scratch reproduces the PDF deterministically: the recommender is fully stateless, the seven sample CSVs are version-controlled, and tie-breaking is documented in the v2 methodology.

4. References

- **Methodology:** `docs/team4_recommender_v2_methodology.md` — the authoritative specification this implementation traces to.
- **EDA v1:** `docs/eda_v1.pdf` — Pangasinan education, employment, and access profile.
- **EDA v2:** `docs/eda_v2.pdf` — Pozorrubio program access, scholarship analysis, and employment signals.
- Burke, R. (2002). Hybrid recommender systems. *User Modeling and User-Adapted Interaction*, 12(4).
- Liu, T.-Y. (2009). Learning to rank for information retrieval. *Foundations and Trends in IR*, 3(3).
- Philippine Institute for Development Studies (2019). *Tracer study of CHED-scholarship and non-scholarship graduates* (Discussion Paper No. 2019-26).